Low-Light Image and Video Enhancement with an Intelligent Assistant (Jarvis)

Project Report

Course: Image and Video Processing

Project Title	Low-Light Image and Video Enhancement with an Intelligent Assistant (Jarvis)
Course	Image and Video Processing
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1. Abstract

Photographs and videos taken in poor light are dark, noisy and hard to use. This project builds a complete low-light enhancement system that works on both images and videos. The program first measures each input (brightness, contrast, noise and entropy), classifies it as Normal, Dim, Dark or Very dark, and then selects an enhancement method automatically. Gamma correction, histogram equalization, CLAHE and Retinex (SSR and MSR) were implemented and compared. Because these classical methods produce blocky colour patches and amplified noise on almost-black images, a new method called Natural (smooth night) was developed. It averages away rounding noise in dark regions, applies an adaptive gamma curve and takes colour from a blurred copy. For video, settings are smoothed across frames to reduce flicker. On synthetically darkened test images, the proposed method kept noise below 1.0 (against 8 to 19 for Retinex and histogram equalization) and gave the best structural similarity on severe darkness (SSIM 0.65 against 0.44 to 0.53). The work is packaged as a desktop and web application with a voice assistant, Jarvis.

2. Introduction

2.1 Background of the problem

Cameras need light. When a scene is dim, the sensor collects few photons, so the picture is dark, low in contrast and noisy. Phones, CCTV cameras, dash cams and drones all produce such images at night or indoors. A very dark photograph is also stored with only a handful of distinct brightness levels, because an 8-bit file records dark areas with values such as 0, 1, 2 and 3. This matters later, when the picture is brightened.

2.2 Why the problem is relevant

Dark images hide information that people and computer programs need. A security officer cannot read a number plate, a doctor cannot see a faint structure, and an object detector loses accuracy when the input is dark. Better-lit images are also simply more pleasant to look at and share.

2.3 Role of image and video processing

Image processing can recover much of the lost information without new hardware. Point operations such as gamma correction change brightness; histogram methods spread the intensity values; Retinex theory separates illumination from reflectance; and filtering reduces noise. For video, the same operations must also remain consistent from frame to frame, otherwise the result flickers.

2.4 Motivation

Low-light enhancement is a practical topic that touches almost every part of the course: histograms, spatial filtering, colour spaces, frequency of noise, quality metrics and video. It also leads naturally to a complete, usable application. I chose it because it can be demonstrated live on any photograph, and because the same code can be shared as a portfolio project. A voice assistant was added so that the tool can be used by simply speaking to it.

3. Problem Statement and Objectives

Problem statement. Images and videos captured in low light are dark, low in contrast and noisy, and existing one-size-fits-all methods either leave them dim or turn very dark photos into blocky, grainy pictures. The project develops a system that analyses any given image or video, chooses a suitable enhancement automatically and produces a clean, natural-looking result.

Objectives

- To measure image darkness objectively using brightness, contrast, noise and entropy, and to classify inputs into four darkness levels.
- To implement and compare classical enhancement methods: gamma correction, histogram equalization, CLAHE, Retinex SSR and Retinex MSR.
- To design a new method (Natural, smooth night) that avoids the blocky artefacts and noise amplification of the classical methods on very dark inputs.
- To extend enhancement to video with frame-to-frame smoothing and to measure flicker before and after.
- To deliver the work as a desktop and web application that includes an assistant, Jarvis, which can scan, enhance and explain results by text and voice.

4. Scope of the Project

4.1 What the project covers

- Analysis of still images and short videos taken in low light, with automatic method selection.
- Eight selectable enhancement methods, a side-by-side comparison of all methods, and before/after quality numbers.
- Video enhancement (first 10 seconds, up to 640 px wide) with temporal smoothing and an ffmpeg re-encode so that the result plays in a browser.
- A Gradio application that runs on a desktop, in Google Colab with a public link, or on Hugging Face Spaces as a website.
- Jarvis: an assistant that works offline with built-in rules, or as a conversational AI when the user supplies a free API key. It listens and speaks through the browser.

4.2 What it does not cover

- It does not use deep learning enhancement models (such as Zero-DCE or RetinexNet). Only classical, explainable techniques are used.
- It cannot recover detail that the camera never recorded; a nearly black photo can be made smooth and visible, but not sharp.
- It does not process very long or very high-resolution videos in real time on ordinary hardware.
- Jarvis cannot see what is inside a photo; he reasons only from the measurements the tools return.

4.3 Type of images and videos used

Colour (RGB) photographs of everyday scenes (portrait, indoor still life, close-up, night landscape) and a short colour video, all darkened to different levels. Any user photo or video can also be loaded into the application.

4.4 Application domain

Night photography, surveillance and security video, dash cameras, mobile photography, and as a pre-processing step before object detection or recognition.

5. Dataset / Input Description

Item	Description
Source of images	Four standard test photographs shipped with the scikit-image library: astronaut, coffee, chelsea (cat) and rocket. The Colab notebook can also load the LOL low-light dataset; results in this report use the sample set only.
Number of images	4 reference images, each darkened at 2 levels, giving 8 dark inputs. Every input was processed by 7 methods (56 enhanced outputs).
Resolution	Reduced to at most 512 pixels on the longer side (for example 512 x 512 for astronaut, about 512 x 341 for coffee and chelsea).
Format	RGB, 8 bits per channel; read and written as PNG/JPEG; handled in OpenCV as BGR uint8 arrays.
Video	One test video, 480 x 270 pixels, 30 frames per second, 3 seconds (90 frames), darkened; MP4 (H.264).
How darkness was created	Moderate: output = 0.6 x (original/255)^2.0 plus Gaussian noise (sigma 1.5). Severe: output = 0.35 x (original/255)^2.6 plus noise (sigma 2.0). The values were rounded to 8 bits, exactly as a real camera file would be.

Because the original photographs are known, each enhanced result can be compared against the ground truth with PSNR and SSIM. This is the main reason for using synthetically darkened images.



Figure 1. Sample input: the coffee image at the severe darkness level.

6. Methodology / Proposed Approach

The pipeline has four stages: (1) analysis, (2) decision, (3) enhancement and (4) evaluation and reporting. Videos follow the same path frame by frame, with the chosen settings smoothed over time.

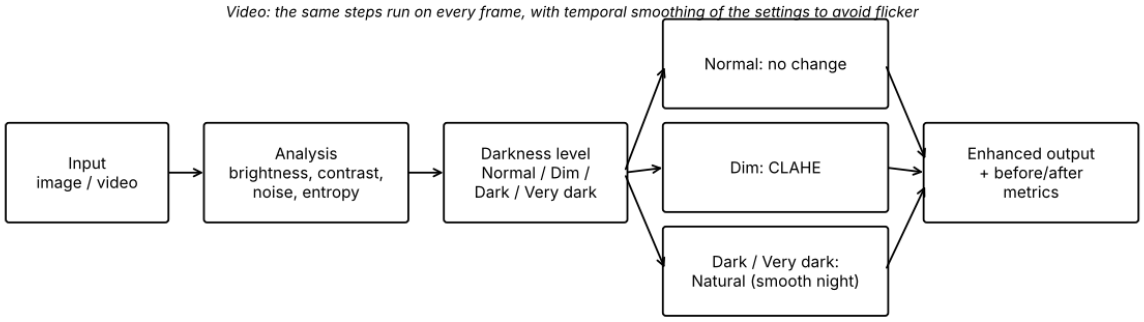


Figure 2. Processing pipeline of the proposed system.

Algorithm (pseudocode)

```
INPUT: image I (BGR, 8-bit)
1. Y = grayscale(I) / 255
2. brightness = mean(Y); contrast = std(Y); noise = Immerkaer(Y); entropy = H(histogram)
3. level = Normal if brightness >= 0.40; Dim if >= 0.25; Dark if >= 0.12; else Very dark
4. if level == Normal: output = I
   elif level == Dim: output = CLAHE(L channel of LAB, clip 3.0, tile 8)
   else: output = NATURAL(I)
5. NATURAL(I):
     y_light = Gaussian(Y, small);  y_heavy = Gaussian(Y, large)
     w = smoothstep(y_heavy)                 # 0 = dark area, 1 = detail
     Ys = w*y_light + (1-w)*y_heavy
     g = log(0.30) / log(mean(Ys))           # mean brightness -> 0.30
     Yt = Ys ** g
     colour = Gaussian(I, large);  ratio = ((colour+e)/(Y_colour+e)) ** 0.85
     output = Yt * ratio
6. report brightness, contrast, entropy, noise before and after
VIDEO: per frame; g = a*g_prev + (1-a)*g_now; blend with previous frame
```

7. Techniques / Algorithms Used

7.1 Image processing

Technique	How it is used in this project
Grayscale conversion	Luminance Y = 0.299 R + 0.587 G + 0.114 B is used for all measurements.
Gamma correction	Output = input^g with g chosen automatically so the mean brightness reaches a target. Fast, but amplifies noise in dark areas.
Histogram equalization	Applied to the luminance channel so colours are not distorted. Spreads intensities strongly but makes noise very visible.
CLAHE	Contrast-limited adaptive histogram equalization on the L channel of the LAB colour space (clip limit 3.0, 8 x 8 tiles). Good for dim images.
Retinex SSR / MSR	Log of the image minus log of its Gaussian-blurred version (illumination estimate); MSR averages three scales (15, 80, 250). Strong shadow recovery, but also strong noise.
Filtering	Gaussian blur (also a fast down-sampled version for large sigmas) is used to estimate illumination and to remove rounding noise.
Colour space conversion	BGR to LAB for CLAHE; luminance/colour-ratio separation for the Natural method.
Feature measurement	Brightness (mean), contrast (standard deviation), entropy (histogram) and noise (Immerkaer's Laplacian-based estimator).
Proposed: Natural (smooth night)	Adaptive blur in dark regions, automatic gamma, and colour from a heavily blurred copy; described in Section 6.

7.2 Video processing

- **Frame extraction:** frames are read one by one with OpenCV and resized to at most 640 px wide.
- **Per-frame enhancement:** the method chosen for the clip (Auto uses the first frames) is applied to every frame.
- **Temporal smoothing:** the gamma value and the Retinex limits are updated by an exponential moving average, and each output frame is blended with the previous one, which reduces flicker.
- **Flicker measurement:** average change in frame brightness between consecutive frames, divided by the mean brightness, in percent.
- **Re-encoding:** the result is re-encoded with ffmpeg to H.264 so that it plays in any browser.

Optical flow, background subtraction and object tracking were not needed because the goal is enhancement, not motion analysis.

8. Implementation

Item	Details
Language	Python 3.10 or newer (tested on 3.10 and 3.14)
Libraries	OpenCV (image and video operations), NumPy (array maths), scikit-image (SSIM, PSNR, test images), Matplotlib (figures), Gradio (web and desktop interface), ffmpeg (video re-encoding)
Tools	Google Colab notebook for analysis and experiments; Gradio app (single file, lowlight_app.py); Hugging Face Spaces and Colab share links for the website
Hardware	Ordinary CPU only; no GPU required. Measured speed: about 100 frames per second for Natural on 480 x 270 video.
Assistant	Jarvis uses browser speech recognition (microphone) and speech synthesis (voice). With a free API key (Google Gemini, Groq or Anthropic) he becomes a conversational AI that can call tools to scan, enhance, brighten, darken, compare or undo the current image. Without a key he uses built-in rules.

Important implementation steps

- Write the measurement functions (brightness, contrast, noise, entropy) and the darkness classifier.
- Implement each classical method so that every function takes a BGR image and returns a BGR image, which lets them be compared with the same code.
- Test the methods on a very dark photo, observe the blocky artefacts, trace them to the few brightness levels in the file, and design the Natural method.
- Add the video enhancer with temporal smoothing, the flicker metric and ffmpeg re-encoding.
- Build the Gradio interface with three tabs (Jarvis, Compare all methods, Video) and connect Jarvis with tool calling.
- Test with automated scripts (60 assistant checks, browser checks with a simulated microphone and voice) and package for desktop, Colab and Hugging Face.

9. Results and Discussion

9.1 Input versus output images

Figures 3 and 4 show the original photograph, the dark input and the result of three classical methods and of the proposed method. Histogram equalization over-brightens and makes noise very obvious (the rocket sky is full of grain). CLAHE is conservative and leaves very dark images dark. Retinex MSR recovers shadows and colour but amplifies noise and gives a harsh, grainy look. The Natural method gives a smooth, clean picture with plausible colour, at the price of some softness.

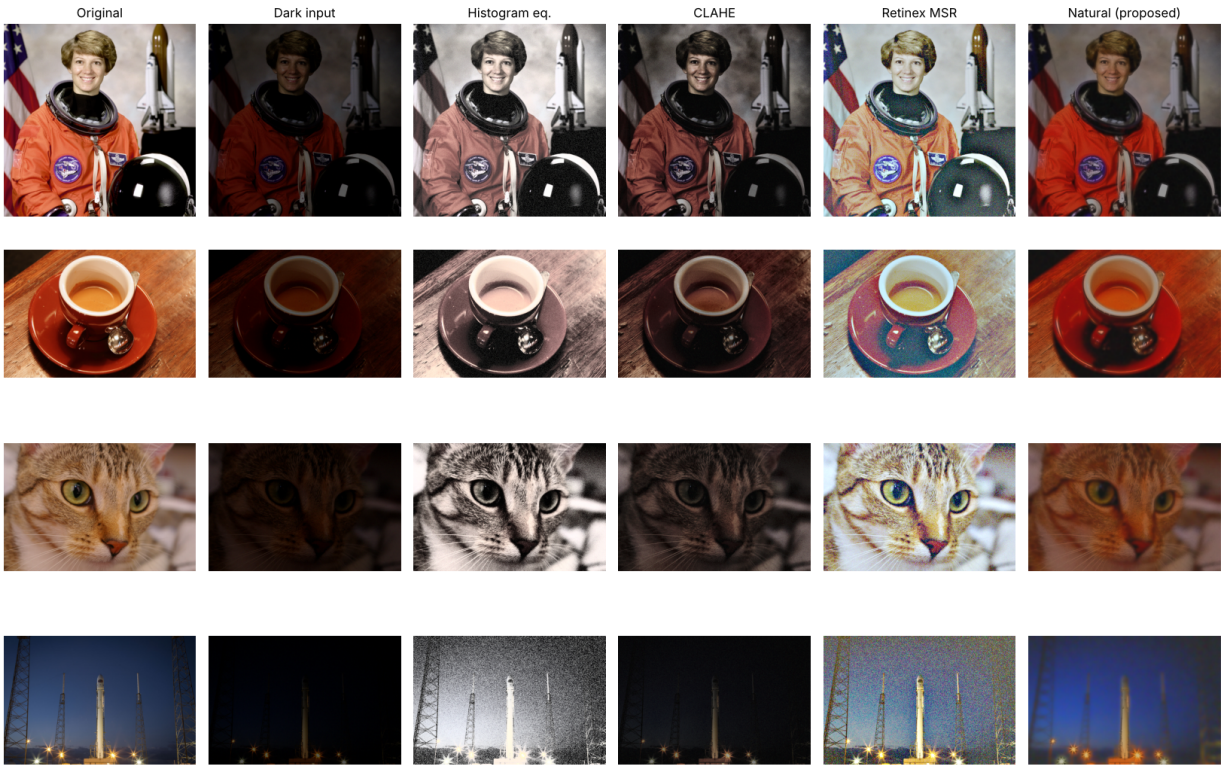


Figure 3. Severe darkness: original, dark input, histogram equalization, CLAHE, Retinex MSR, and the proposed Natural method.

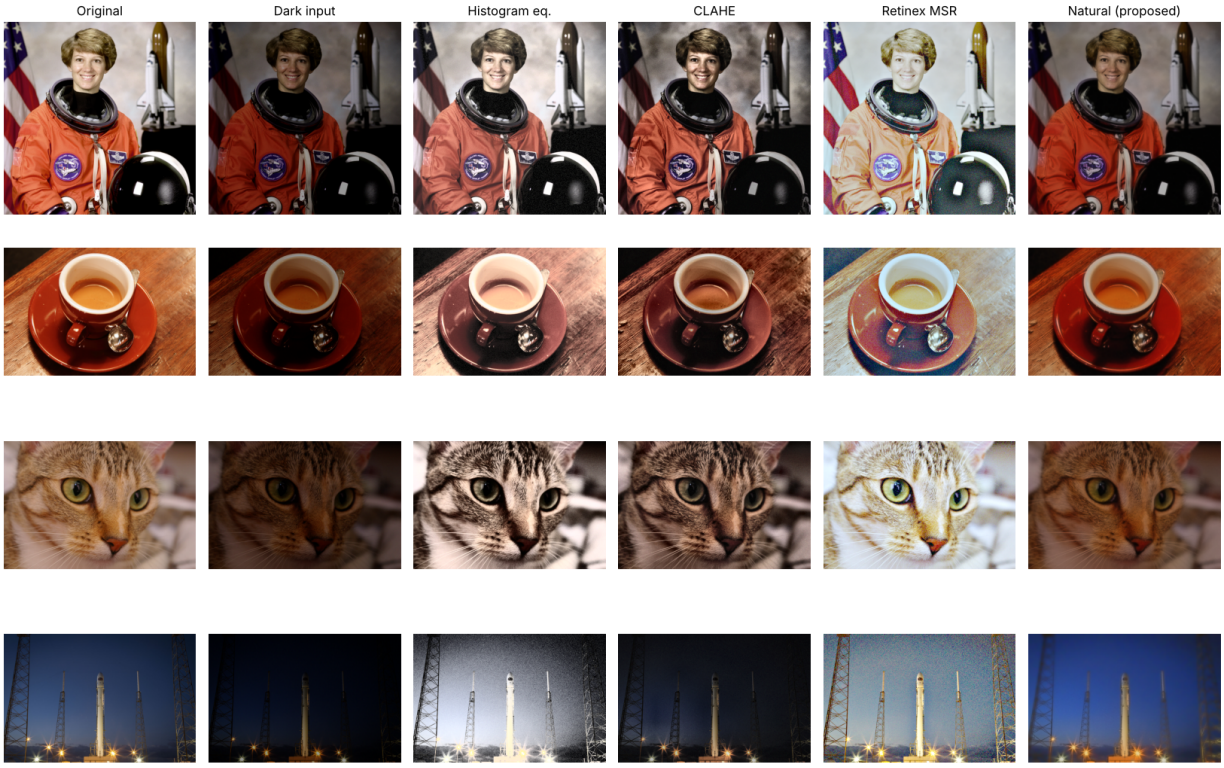


Figure 4. Moderate darkness: the same comparison.

9.2 Intermediate result: histograms

The histogram of the dark input is squeezed into the lowest few bins. After enhancement the intensities are spread over a wider range, which shows the recovery of contrast, while the shape remains smooth because the rounding steps were averaged away first.

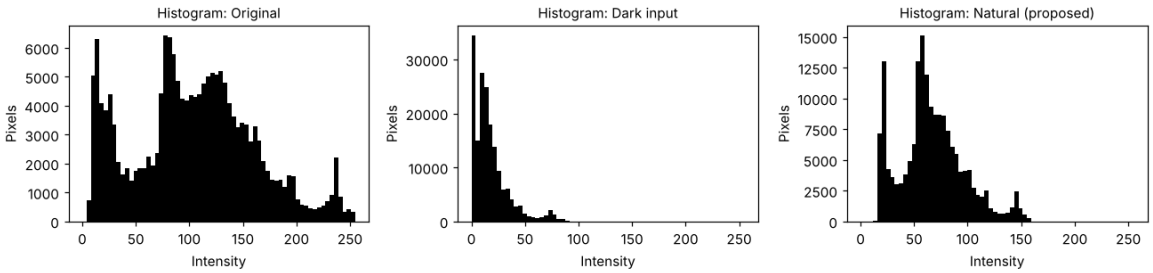


Figure 5. Intensity histograms of the original, dark and enhanced coffee image (severe level).

9.3 Application screenshot

Figure 6 shows the application after a dark photo was uploaded. Jarvis scans the image, names the darkness level, chooses the method, enhances it and states the change in brightness. The table under the result gives the before and after measurements.

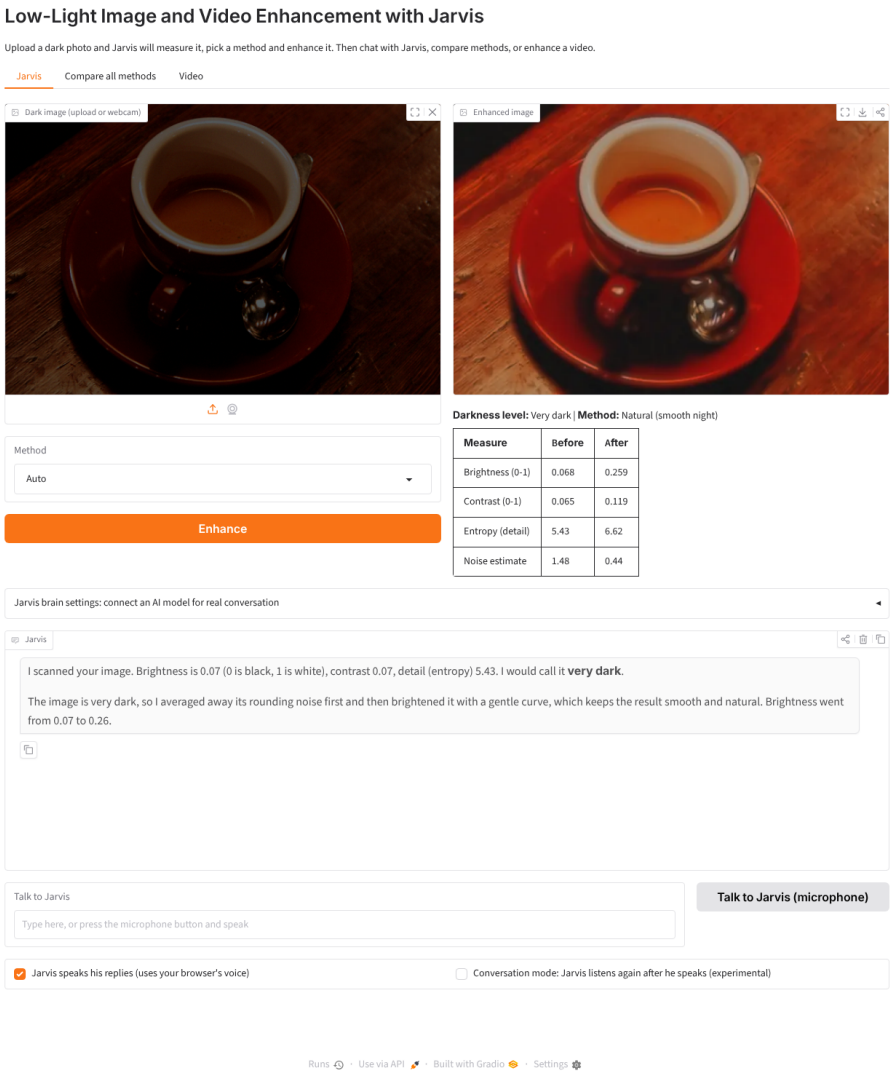


Figure 6. The Jarvis tab: input, enhanced output, before/after table and the assistant's message.

9.4 Quantitative performance

Each value is the mean over the four test images. PSNR and SSIM compare the result with the original photograph (higher is better). Entropy measures detail. Noise is Immerkaer's estimate (lower is cleaner). The ideal brightness of a well-exposed photo is around 0.3 to 0.5.

Table 1. Moderate darkness (mean of 4 images).

Method	PSNR (dB)	SSIM	Entropy	Brightness	Noise	Time (ms)
Input (no enhancement)	10.98	0.425	5.95	0.130	1.73	-
Gamma	20.41	0.757	7.04	0.389	5.07	4
Histogram equalization	15.54	0.589	6.72	0.497	8.93	1
CLAHE	15.28	0.661	6.89	0.255	5.06	29
Retinex SSR	12.39	0.601	7.31	0.589	8.13	17
Retinex MSR	12.00	0.591	7.32	0.604	8.64	35
Natural (proposed; used by Auto)	16.39	0.755	6.72	0.273	0.95	15

Table 2. Severe darkness (mean of 4 images).

Method	PSNR (dB)	SSIM	Entropy	Brightness	Noise	Time (ms)
Input (no enhancement)	8.71	0.193	4.84	0.058	1.50	-
Gamma	18.25	0.525	6.89	0.339	11.83	5
Histogram equalization	14.02	0.453	5.70	0.496	19.40	1
CLAHE	11.02	0.460	5.89	0.145	4.52	2
Retinex SSR	12.98	0.452	7.47	0.547	15.10	16
Retinex MSR	12.82	0.440	7.48	0.553	16.57	33
Natural (proposed; used by Auto)	15.88	0.652	6.33	0.264	0.81	12

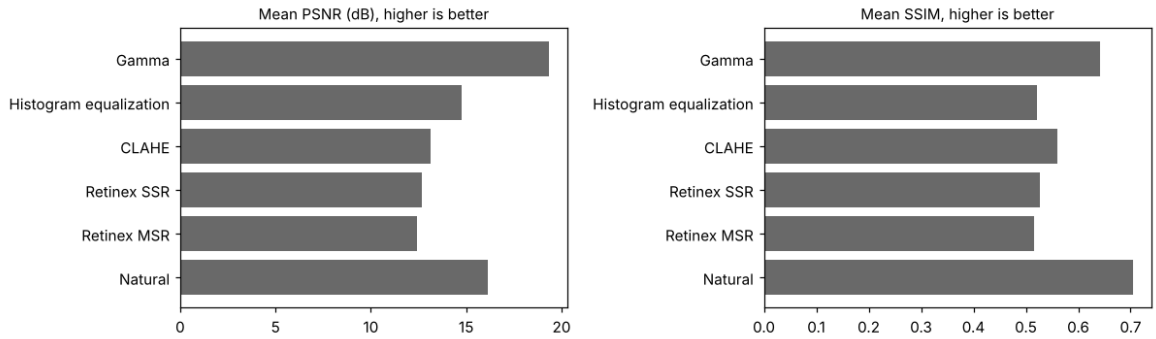


Figure 7. Mean PSNR and SSIM of each method over both darkness levels.

9.5 Discussion of the numbers

- **Noise.** The proposed method has by far the lowest noise (about 0.8 to 1.0), whereas histogram equalization and Retinex raise it to 8 to 19 and 8 to 17. This is the main reason the results look clean.
- **Structure.** At severe darkness, the proposed method has the highest SSIM among the enhancement methods that also fix brightness (0.652, against 0.440 to 0.460 for histogram equalization, CLAHE and Retinex), and it improves SSIM from 0.193 for the raw dark input.
- **Where gamma wins.** Plain gamma has a higher mean PSNR (20.4 dB moderate, 18.3 dB severe). This should be read with care: the test images were darkened with a power law, so a power-law correction partly undoes exactly that operation. Gamma also leaves the noise at 5 to 12, and on the very dark rocket image its SSIM is only 0.25 against 0.76 for the proposed method.
- **Detail.** The entropy of the proposed method (6.3 to 6.7) is lower than that of Retinex (7.3 to 7.5). Retinex's higher entropy is partly noise being counted as detail.
- **Speed.** All methods take between 1 and 35 ms for a 512-pixel image on a CPU; the Natural method takes about 12 to 15 ms.

9.6 Video results

A 3-second dark video (90 frames, 480 x 270) was processed with three methods. Flicker is the average frame-to-frame change in brightness relative to the mean brightness.

Method	Brightness before	Brightness after	Flicker before	Flicker after	Speed (frames/s)
Auto (Natural)	0.092	0.250	15.81%	6.73%	99
CLAHE	0.092	0.207	15.81%	10.08%	220
Gamma	0.092	0.337	15.81%	4.85%	160

The enhanced videos are brighter and the temporal smoothing cut flicker to less than half of the original value for the Auto method. Plain gamma showed the lowest flicker because it does not adapt to the content, but it leaves more noise. The Natural method ran at about 99 frames per second, which is faster than real time for this resolution.



Figure 8. A video frame before (left) and after (right) enhancement with the proposed method.

10. Challenges / Limitations

Problems encountered

- **Blocky patches on very dark photos.** A real very dark photo (a moon photograph) came out with coloured blocks after Retinex. The cause is that the file holds only a few brightness levels, and brightening stretches the rounding steps into visible contours. Averaging in dark areas before brightening, and taking colour from a blurred copy, fixed this and led to the Natural method.
- **Halos.** The first version of the Natural method produced a glow around dark objects on moderately dark images. The blur size is now adapted to how dark the image is.

Difficult cases and failures

- Photos that are almost black cannot be fully recovered; the result is smooth but soft, and a faint colour tint may remain around bright regions such as clouds near the moon.
- Scenes with strong light sources (street lamps, rocket flame) can show a slight glow because the blur spreads bright values.
- On the test set, the proposed method has a lower PSNR than plain gamma, and a lower entropy than Retinex, so it favours cleanliness over maximum sharpness.

Computational and dataset limitations

- Video is limited to the first 10 seconds and 640 px wide to keep the free web version responsive.
- The quantitative test used synthetic darkening of four photographs, not images from real low-light cameras with paired ground truth. A study on the LOL dataset would be the next step.
- Jarvis's conversation quality depends on an external AI service and the free quota of the user's key; without it he falls back to built-in rules.

11. Conclusion

What was implemented. A complete low-light enhancement system for images and videos, with objective darkness analysis, five classical methods, a new Natural (smooth night) method, temporal smoothing for video, a Gradio desktop and web application, and the Jarvis assistant that can listen, speak and use the tools.

Techniques used. Brightness, contrast, entropy and noise measurement; gamma correction; histogram equalization; CLAHE; Retinex SSR and MSR; Gaussian filtering; colour-space conversion; exponential smoothing and frame blending for video; PSNR and SSIM for evaluation.

What was achieved. On the test set the proposed method produced clean pictures with almost no added noise and the best structural similarity at severe darkness, and reduced video flicker to less than half. The system chooses its method automatically and reports its measurements.

Learning. The most important lesson was that numbers and pictures must be examined together: a method with a good average score can still look wrong on a real photo, and an artefact must be traced to its cause (here, the limited brightness levels of a dark file) before it can be fixed. I also learned how to measure image quality, how to avoid flicker in video and how to turn a set of algorithms into an application that other people can use.

12. References

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